

* Define current sources and voltage sources.

→ A current source is a source that supplies constant current regardless of voltage.

A voltage source is a source that supplies ^{constant} voltage regardless of voltage.

* Define biasing of a BJT.

→ Biasing a BJT is the process of applying suitable DC voltages and currents to establish a stable operating point for proper functioning.

* Define FET and write its two applications.
→ A FET is a voltage controlled semiconductor device in which the current is controlled by an electrical field. It is commonly used in amplifiers and switching circuits.

* Write the Barkhausen criteria.

→ The Barkhausen criteria states that for sustained oscillations, the total loop gain of the system must be equal to unity and the total phase shift around the loop must be 0° or 360° .

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* Define ideal operational amplifier

→ An ideal operational amplifier is a device that has infinite gain, infinite input impedance, zero output impedance and infinite bandwidth and no offset voltage.

* State Kirchhoff's first and second law.

→ Kirchhoff's first law states that the sum of currents acting at a node is 0.

Kirchhoff's second law states that the algebraic sum of voltages in a closed loop is 0.

* What is ~~a photo~~ the difference between photodiode and led?

→ A photodiode converts light energy into electrical energy and a LED converts electrical energy into light.

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* What are ~~the~~ ' α ' and ' β ' parameters and what is their relation.

→ α -parameter refers to the ratio of collector current to emitter current while β is the ratio of collector current to base current.

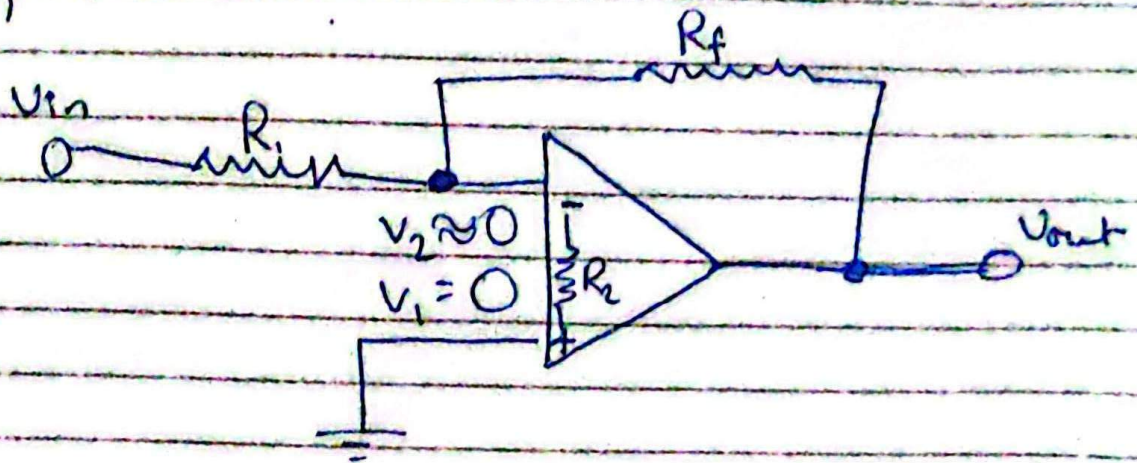
$$\alpha = \frac{\beta}{1 + \beta}$$

* What do you mean by JFET? How does it differ from ordinary transistors?

→ A JFET is a voltage controlled device in which current is controlled by the electric field at the gate terminal. It differs from ordinary transistors as it is voltage controlled and has high input impedance.

* What do you mean by virtual ground in an operational amplifier.

→ Virtual ground is a point in an op. amp. circuit that is at 0 potential but is not physically connected to the ground. It occurs due to high ~~imp~~ gain of the amplifier and negative feedback.



* State the condition under which feedback amplifier works as an oscillator.

→ A feedback amplifier works as an oscillator when the feedback is positive and Barkhausen criteria is met.

* Distinguish between regulated and unregulated power supply.

→ A regulated power supply provides a constant output voltage despite variation in input or load conditions, whereas an unregulated power supply produces an output that varies with changes in input voltage or load.

* What is current booster?

→ A current booster is a circuit usually involving a transistor, that increases the current supplying capability of a system without significantly changing the voltage.

* Define pinch-off voltage.

Pinch off voltage refers to the value of drain to source voltage at which the current through a FET becomes constant, and the channel is effectively closed.

* Define DC load line and Q-Point

→ The DC load line is a graphical representation of all possible combinations of collector current and collector-emitter voltage.

The Q-point is the operating point, is the operating point on this line where the transistor operates under no signal condition.

* Differentiate between BJT and FET

→ BJT is current controlled, bipolar, low input impedance device, whereas FET is voltage-controlled, unipolar device with high input impedance.

* Differentiate between ~~MOSFET~~ MOSFET and JFET

→ A MOSFET has an insulated gate and can operate in both enhancement and depletion mode. Whereas a JFET has a PN junction gate and only operates in depletion mode.